



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES**

**B.Sc. (General) Degree in Applied Sciences
First year – Semester I Examination – May/June 2017**

FDN 1203 – GENERAL BIOLOGY

Time: Two (02) hours

Part A – Multiple choice questions.

Answer all questions. Underline the correct option on the script itself.

PART A

1. What is the pulse rate of an average healthy human being?
 - (a) 65 beats per minute
 - (b) 100 beats per minute
 - (c) 16 beats per minute
 - (d) 72 beats per minute

2. Among the biotic components of the ecosystem the producer system is,
 - (a) rivers
 - (b) sea
 - (c) soil
 - (d) green plants

3. Which of the following organisms do not obey 'cell theory'?
 - (a) Virus
 - (b) Bacteria
 - (c) Fungi
 - (d) Archea

4. The movement of molecules against concentration gradient is called,
 - (a) osmosis
 - (b) active transport
 - (c) diffusion
 - (d) passive transport

5. Prokaryotic cells lack,
- (a) nucleolus
 - (b) nuclear membrane
 - (c) membrane bound organelles
 - (d) all of the above
6. Which of the following, correctly matches an organelle with its function?
- (a) Mitochondrion - Photosynthesis
 - (b) Nucleus - Cellular respiration
 - (c) Central vacuole - Storage
 - (d) Ribosome - Manufacture of lipids
7. The sequence of energy flow through a food chain is,
- (a) Primary consumers → Producers → Higher order consumers
 - (b) Producers → Primary consumers → Higher order consumers
 - (c) Primary consumers → Higher order consumers → Producers
 - (d) Producers → Higher order consumers → Primary consumers
8. Which of the following is a characteristic of parasitism?
- (a) One organism kills and consumes another
 - (b) Two organisms live together and neither is harmed
 - (c) Two organisms nourish each other and both benefit
 - (d) One organism lives in or on another and gains benefit
9. The phenomenon in which high levels of carbon dioxide cause global warming is known as,
- (a) the Gaia hypothesis
 - (b) acid deposition
 - (c) the greenhouse effect
 - (d) global distillation
10. Which of the following nitrogen bases form Hydrogen bonds with Thymine and Guanine respectively, in DNA?
- (a) Adenine, Cytosine
 - (b) Guanine, Cytosine
 - (c) Thymine, Guanine
 - (d) Cytosine, Adenine

11. A lake rich in nutrients and organisms is classified as,
- (a) dystrophic
 - (b) oligotrophic
 - (c) ecotrophic
 - (d) eutrophic
12. When subunits called monomers are bonded together they form large molecules called polymers. Carbohydrate is a type of polymer. What's the monomer that makes up carbohydrates?
- (a) Amino acid
 - (b) Ion
 - (c) Monosaccharide
 - (d) Nucleotide

13. Assertion : A cell membrane shows fluid behaviour
Reason : A membrane is mosaic or composite of diverse lipids and proteins.

Considering the above statements,

- (a) Assertion is true but the reason is false
 - (b) Both the assertion and the reason are false
 - (c) Both the assertion and the reason are true
 - (d) The reason is a true statement but it doesn't explain the assertion
14. Night blindness is caused by the lack of,
- (a) vitamin A
 - (b) vitamin B
 - (c) vitamin C
 - (d) vitamin D

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15. Which of the following is not a part of the scientific process?
- (a) Making predictions
 - (b) Asking questions
 - (c) Using creative insights
 - (d) Proving theories are true
16. are transported by a particular carrier or passed through a particular channel.
- (a) All water soluble molecules
 - (b) Certain water soluble molecules
 - (c) All insoluble molecules
 - (d) Only the smallest molecules

17. Which of the following statements about symbiotic relationship is true?
- (a) In a parasitic relationship, both organisms are harmed
 - (b) Symbiotic organisms have usually undergone little or no co evolution
 - (c) The most efficient type of parasite is one that kills its host
 - (d) None of the above stands true
18. The best predictor of the future population growth rate is,
- (a) life expectancy
 - (b) age structure
 - (c) mortality
 - (d) undeveloped land
19. Which of the following strategies is least likely to improve the world's food supply?
- (a) Utilizing new foods
 - (b) Bringing more land under cultivation
 - (c) Using genetic engineering to improve crops
 - (d) None of the above
20. Due to the rising levels of CO_2 in our atmosphere, what would be the incident that is **NOT** likely to happen?
- (a) Increased incidence of skin cancer
 - (b) Rise in sea level
 - (c) Shifts in locations of deserts and fertile regions
 - (d) Change of climatic patterns

(1x20=20 marks)

Part B-Structured essay question.**Answer all (two) questions. Use the provided spaces on the script itself.****PART B**

1. a) State the eight (08) characteristics of life.

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(04 marks)

- b) Briefly discuss how homeostasis is maintained in an organism.

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(06 marks)

2. a) What is meant by 'bio-geo chemical cycles'?

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(04 marks)

b) Concisely review the process of water/hydrologic cycle.

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(10 marks)

c) Write four (04) steps you can implement to stop “global warming”.

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(04 marks)

Part C – Essay Questions.

Answer all(two) questions using diagrams where appropriate. Use the booklet provided.

PART C

1. a) Write the elements of the 'Cell Theory'.

(02 marks)

- b) Compare a bacterial cell with an animal cell.

(05 marks)

- c) Discuss the structure of the following cell components associated to its function.

- i. Cell membrane
- ii. Endoplasmic reticulum
- iii. Lysosomes

(15 marks)

2. a) Using an example for each, explain the following ecological concepts.

- i. Tropic levels and energy flowing
- ii. Prey-predator relationship
- iii. Exponential and logistic growth curves
- iv. Carrying capacity

(20 marks)

- b) Write a brief account on "Acid Rain".

(10 marks)

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